

TECHNOLOGY AT YOUR SERVICE

Chinese university unveils mind-controlled car

A mind-controlled car that would do away with the use of steering wheel or accelerator has been unveiled in China. It has been developed by a research



Mind-controlled car by Nankai University in Tianjin (Image courtesy: www.huffingtonpost.co.uk)

team from Nankai University in Tianjin in collaboration with Chinese car maker Great Wall Motor.

The car is controlled via a headset with 16 sensors that send impulses from the user's brain to the car's processing system. Sensors capture brain signals and the recognition system analyses these, translates these into driving instructions and sends these to the car.

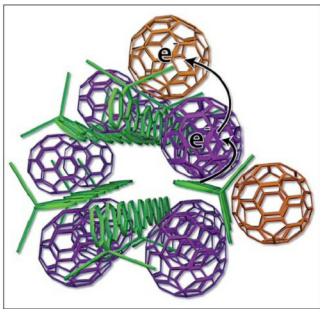
Plant-inspired solar cells revolutionise energy storage

Technology developed by scientists at University of California, Los Angeles (UCLA), could store solar energy for up to several weeks.

Most current residential rooftop solar panels can store energy from the sun for only a few microseconds at a time. The new design is inspired by the way plants generate energy through photosynthesis.

The two components include a polymer donor and a nano-scale fullerene acceptor. The polymer donor absorbs sunlight and passes electrons to the fullerene acceptor. The process generates electrical energy.

UCLA technology arranges the elements neatly like small bundles of uncooked spaghetti with precisely placed meatballs. Some meatballs are designed to sit inside the spaghetti bundles, but others are forced to stay on the outside. Meatballs inside the structure take electrons from the polymers and toss these to the outside, which can effectively keep electrons away from the polymer for weeks. The materials self-assemble just by being placed in close proximity.



Technology by University of California, Los Angeles, could allow storage of solar energy up to several weeks (Image courtesy: <http://newsroom.ucla.edu>)

Memcomputer mimics human brain

As the need for computing grows, researchers are faced with the problem of creating more computing power from existing electronics. Enter memcomputers. Dr Massimiliano Di Ventra, computer scientist at University of California, and his team, is developing a memcomputer, which is a special computer that derives inspiration from human brain.

Basic computers store data in one location and process it at some other (central processing unit, or CPU). The data travels to and fro between these locations for every computing cycle.

Memcomputers would take only a few seconds to solve complex mathematical problems that the current generation of supercomputers takes decades to solve.

Di Ventra explains this with an example. The human brain consumes about 20W to perform about 10^{16} operations per second. A supercomputer, on the other hand, would need 10 million times more power to do so.

Memcomputers would be able to do this with the help of memprocessors, which function like transistors, and are the basic building blocks of modern electronics. But with additional function, these will also store charge, which would allow these to store and process data simultaneously, just like the human brain.

Software to detect eye diseases

Smartphones can detect heart rate and blood-oxygen content. Soon, these would be able to detect common health-related issues. Researchers at Medical and Surgi-